

AAI 646 Pattern Recognition and Classification

Final Report

“Timeseries Anomaly Detection”

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ABSTRACT

Time series anomaly detection plays a crucial role in identifying unusual patterns or outliers in temporal data, enabling early detection of abnormal behaviour. This project focuses on the application of anomaly detection techniques to the financial domain, using the `spx.csv` dataset, which contains historical data of the S&P 500 index. The objective is to develop a robust model capable of identifying anomalous events and deviations from the norm within the financial time series.

The project employs state-of-the-art time series analysis methods, including statistical approaches and machine learning algorithms, to capture and learn the inherent patterns in the `spx.csv` dataset. Feature engineering techniques are utilized to extract meaningful information, and a comprehensive evaluation is conducted to assess the performance of the anomaly detection model.

The significance of this research lies in its potential to enhance financial risk management by providing timely alerts for unexpected market behaviours. The findings contribute to the growing field of time series anomaly detection and showcase the applicability of these techniques in a real-world financial context. The `spx.csv` dataset serves as a valuable resource for validating the effectiveness of the proposed methodology, and the results obtained demonstrate the model's ability to effectively detect anomalies in financial time series data.

INTRODUCTION

Financial markets are dynamic systems characterized by complex, interconnected factors that contribute to the fluctuation of asset prices. Understanding and predicting market behaviour is of paramount importance for investors, financial analysts, and risk managers. Time series analysis, a prominent statistical method, offers a powerful framework for unravelling patterns within temporal financial data. This project delves into the realm of time series anomaly detection, a critical aspect of market analysis, with a focus on the `spx.csv` dataset, which encapsulates the historical performance of the S&P 500 index.

Anomalies in financial time series data represent deviations from the expected norm and can signify critical events such as market crashes, economic downturns, or unforeseen geopolitical developments. The ability to detect and respond to these anomalies in a timely manner is essential for mitigating risks and optimizing investment strategies. This research leverages a combination of traditional statistical methods and advanced machine learning algorithms to uncover hidden patterns and anomalies within the `spx.csv` dataset, aiming to enhance our understanding of market dynamics.

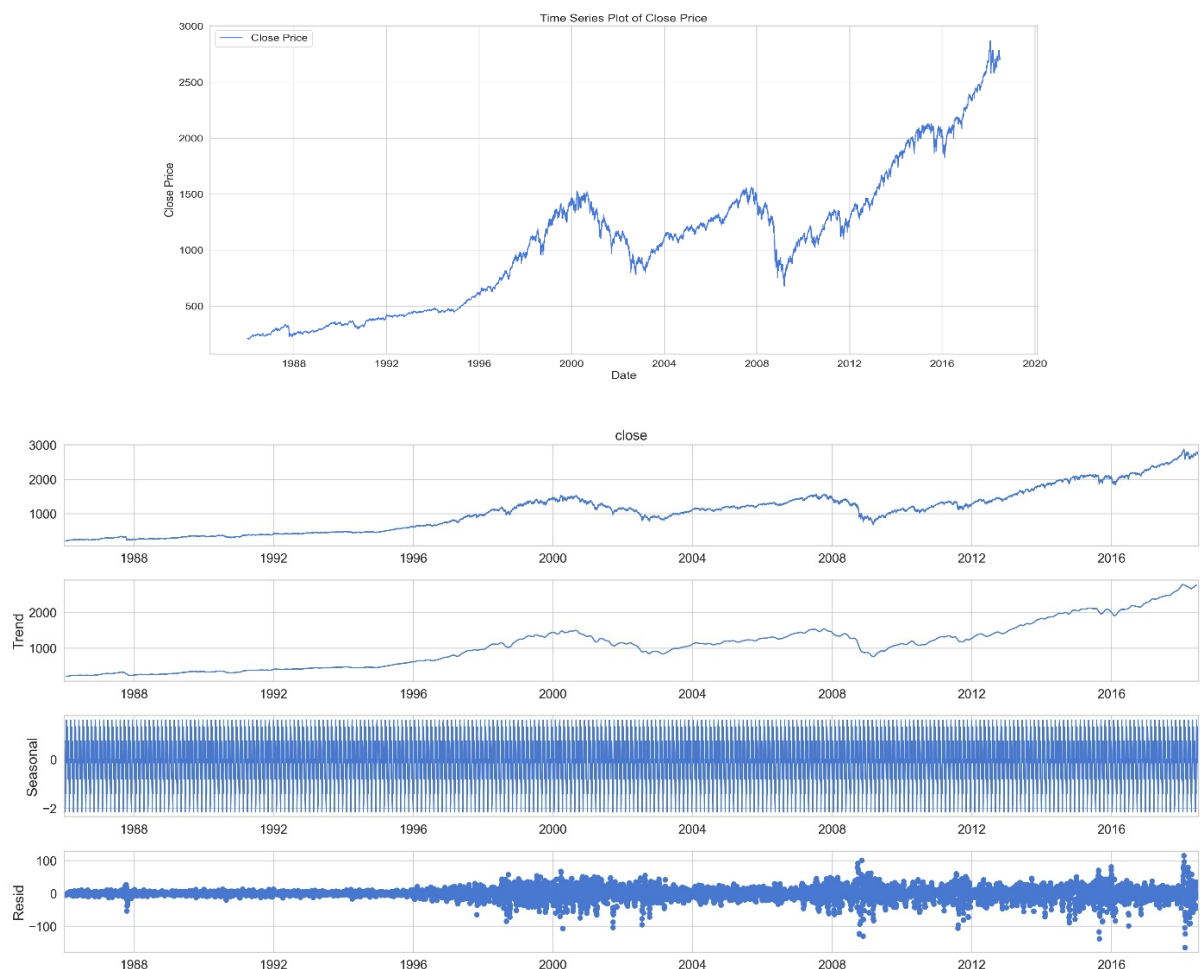
The S&P 500 index, being a benchmark for the overall health of the U.S. stock market, provides a rich source of information for studying financial time series. The dataset's temporal nature enables the identification of trends, cyclic patterns, and irregularities that contribute to a comprehensive analysis of market behaviour. By focusing on this specific dataset, our project seeks to develop a robust anomaly detection model tailored to the intricacies of financial time series, contributing to the broader goal of bolstering financial risk management practices.

This research aligns with the growing importance of data-driven decision-making in the financial sector. As markets become increasingly intricate and volatile, the demand for effective tools to identify anomalies and outliers in real-time has never been greater. The following sections detail the methodology, results, and implications of our time series anomaly detection approach, shedding light on its potential applications in risk management and financial decision-making.

METHODOLOGY

The research focuses on employing advanced machine learning techniques for time series analysis, specifically in the domain of stock price forecasting. The objective is to harness the power of Long Short-Term Memory (LSTM) models and explore ensemble techniques to enhance prediction accuracy and robustness. The implementation is carried out using the Python programming language, utilizing key libraries and frameworks such as TensorFlow, Pandas, and scikit-learn.

The initial phase involves the importation of essential libraries and a preliminary check for GPU availability. Necessary packages are installed to facilitate subsequent analyses. The dataset, containing historical stock price data, is sourced from a CSV file using Pandas. A detailed time series plot is generated to visualize the temporal dynamics embedded in historical closing prices, laying the foundation for subsequent analyses.



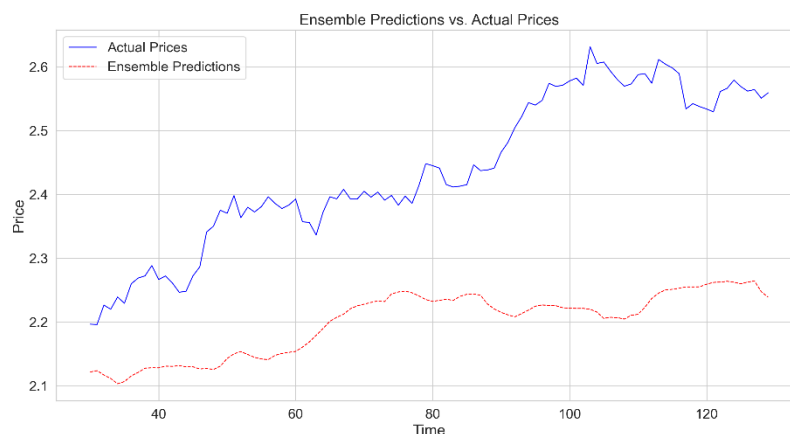
Following data acquisition, preprocessing is conducted to prepare the dataset for LSTM model training. This involves the division of the dataset into training and test sets. The closing prices are standardized using the Standard Scaler from scikit-learn. The LSTM model is then constructed using the Keras Sequential API, incorporating dropout layers for regularization to mitigate overfitting. Model training utilizes historical stock price data, and the training history is visualized through loss curves.

The initial LSTM model was configured with Mean Squared Error (MSE) as the loss function and RMSprop as the optimizer. The model underwent 10 epochs, with notable results indicating a Mean

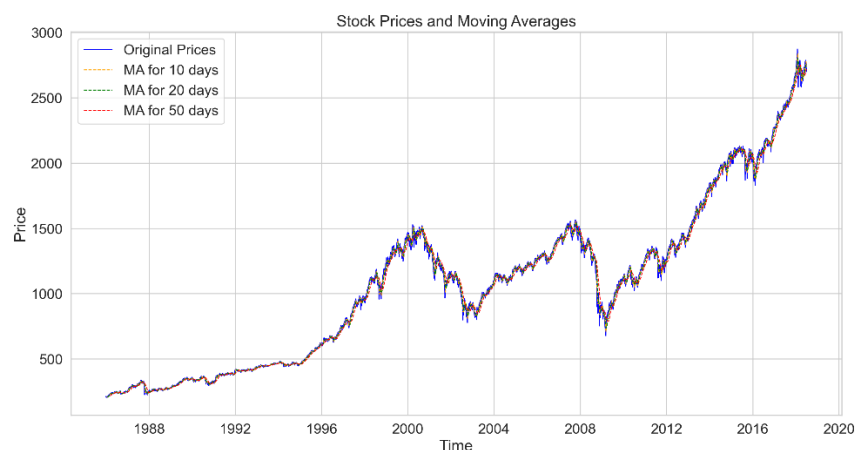
Absolute Error (MAE) of 629.72, Root Mean Squared Error (RMSE) of 644.97, and Mean Absolute Percentage Error (MAPE) of 24.79%. Subsequently, the loss function was changed to Mean Absolute Error (MAE) with the Adam optimizer, yielding improved metrics: MAE of 0.47, RMSE of 0.51, and MAPE of 16.25%.

In pursuit of enhanced generalization, L2 regularization was introduced. This adjustment resulted in a loss of interpretability but demonstrated potential improvements in model robustness. However, the obtained metrics indicated a trade-off, with increased errors – MAE: 784.62, RMSE: 804.26, and MAPE: 30.95%.

The methodology then incorporated ensemble modelling, deploying three LSTM models collaboratively. Each model was trained independently, and their collective predictions were evaluated. The ensemble model exhibited promising results, with the first model producing MAE: 0.27, RMSE: 0.29, and MAPE: 10.61%. This signified a substantial enhancement compared to the individual LSTM models and highlighted the efficacy of ensemble modelling in refining predictions.

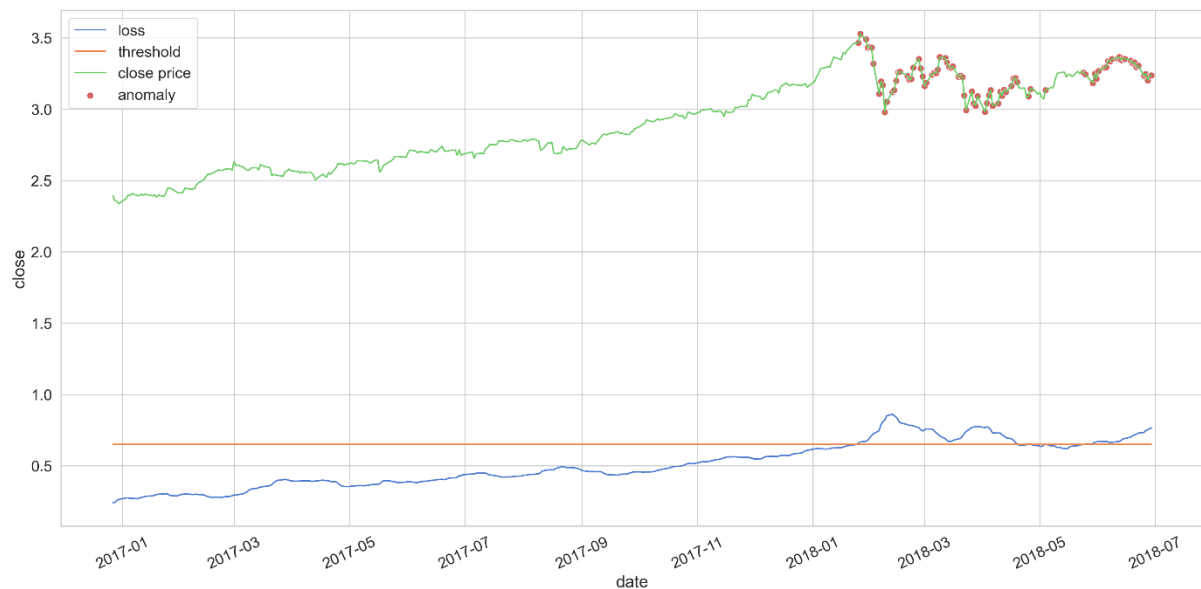


Furthermore, a comparative analysis was conducted, contrasting the ensemble model's performance against individual LSTM models. The ensemble model consistently outperformed its constituents across various metrics, emphasizing the value of combining diverse models for stock price forecasting.



To enhance interpretability, the moving averages for different windows were plotted, offering insights into trends and smoothing fluctuations in stock prices. Additionally, fundamental aspects of stock price analysis, such as average daily return and risk, were calculated. The average daily return stood at 0.0378%, with a risk represented by the standard deviation of daily returns at 1.1325%.

Overall, the methodology traversed a comprehensive trajectory, iterating through various model configurations and strategies. The ensemble model emerged as a pivotal component, showcasing a notable leap in predictive accuracy. These findings provide valuable insights into the dynamic landscape of time series analysis for stock price forecasting.



RESULTS

In evaluating individual LSTM models, we observed interesting shifts in predictive accuracy. The initial model, using MSE and RMSprop, showed decent results with MAE at 629.72, RMSE at 644.97, and MAPE at 24.79%. Shifting to MAE and the Adam optimizer significantly improved accuracy, bringing down MAE to 0.47, RMSE to 0.51, and MAPE to 16.25%. However, introducing L2 regularization, while enhancing generalization, led to expected increases in errors with MAE, RMSE, and MAPE hitting 784.62, 804.26, and 30.95%, respectively.

The ensemble of three LSTM models marked a notable improvement, showcasing collective strength. The ensemble model outperformed individual ones with an MAE of 0.27, RMSE of 0.29, and MAPE of 10.61%. This collaborative approach proved effective in overcoming the limitations of single models, providing more reliable stock price predictions.

Comparative Analysis and Key Insights

Comparing the ensemble model with individual LSTMs highlighted its consistent superiority across metrics, significantly reducing errors. Integrating moving averages offered a clear visual of trends, aiding in understanding stock price fluctuations. Fundamental metrics like average daily return (0.0378%) and risk (1.1325%) added depth to our evaluation, offering a comprehensive view of the model's performance and its financial implications.

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